

Vineeth G.

AI/ML Engineer

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SUMMARY

AI/ML Engineer with 5+ years of experience developing high-impact predictive models, MLOps pipelines, and Agentic AI workflows using Python and AWS. Proven track record of accelerating retraining cycles, engineering low-latency production microservices, and optimizing data retrieval bottlenecks to handle millions of daily transactional records.

TECHNICAL SKILLS

Methodologies: SDLC, Agile, Waterfall

Programming Languages: Python, SQL, R

Machine Learning & MLOps: PyTorch, TensorFlow, Scikit-learn, XGBoost, Pandas, NumPy, MLflow, Feature Engineering, Model Deployment

Database: MySQL, PostgreSQL, Pinecone/Milvus (Vector DBs)

Cloud & DevOps: AWS (SageMaker, S3, EC2), Azure, Docker, FastAPI, Jenkins (CI/CD)

AI Orchestration & Agents: Multi-Agent Systems, Autonomous Agents, Agentic Workflows, LangGraph, LangChain, LlamaIndex, Model Context Protocol (MCP)

Generative AI & LLM Engineering: Large Language Models (LLMs), Advanced RAG Pipelines, Context Window Optimization, Fine-Tuning (LoRA, QLoRA), Prompt Engineering, Quantization

AI Evaluation & Observability: LLM-as-a-Judge, Ragas, DeepEval, LangSmith/Langfuse (Tracing & Cost Optimization)

Soft Skills: Problem-Solving, Collaboration with Cross-functional Teams, Strong Communication, Adaptability

WORK EXPERIENCE

AI/ML Engineer | PNC, Texas

July 2023 – July 2026

- Designed an intelligent cash flow forecasting engine for an enterprise banking platform using **Python** and **TensorFlow**, automating data predictions across millions of records to improve system asset allocation efficiency by **24%**.
- Developed a real-time transactional fraud detection model using **XGBoost** and **Scikit-learn**, processing **50K+** payment requests hourly with sub-100ms inference latency to mitigate unauthorized account activity.
- Implemented an end-to-end **MLOps pipeline** utilizing **Docker**, **Kubernetes**, and **AWS SageMaker** to automate model training and validation, reducing the automated retraining cycle from **8 hours to 45 minutes**.
- Engineered robust data preprocessing and feature selection workflows using **Pandas** and **NumPy** on **2M+ customer transaction records**, applying **SHAP-based feature importance** analysis to improve risk model accuracy by **19%**.
- Integrated **machine learning models** into secure production microservices via **FastAPI REST APIs**, serving real-time risk scoring for 200K+ daily customer requests with sub-150ms latency and zero peak-hour downtime.
- Architected an internal multi-agent orchestration framework using **LangChain** and **Python**, implementing **Agentic AI** workflows to automate customer data validation and reduce manual compliance review times by **30%**.
- Collaborated with data engineering teams to optimize high-performance data schemas using **SQL** and distributed databases, reducing model data-retrieval bottlenecks by **35%** during high-traffic system hours.
- Containerized and deployed predictive customer banking models onto **Red Hat OpenShift** container environments, optimizing cloud infrastructure utilization and reducing system deployment times by **50%**.

Machine Learning Engineer | Vivma Software Inc, India

May 2019 – November 2021

- Designed an automated product recommendation engine for an e-commerce platform using **collaborative filtering** and matrix factorization in **Python**, achieving a **24% improvement** in customer conversion rates.
- Developed a predictive customer lifetime value (CLV) model using **XGBoost** and **Scikit-learn**, processing **50K+ customer profiles hourly** on **AWS EC2** compute instances to help boost marketing campaign revenue by **16%**.
- Implemented automated data pipelines and baseline model tracking setups using **Docker** and **Amazon SageMaker**, streamlining background workflows to reduce the manual model retraining cycle from **8 hours to 45 minutes**.
- Engineered robust data preprocessing and feature selection workflows using **Pandas** and **NumPy** on **2M+ customer transaction records**, optimizing feature arrays to increase behavioral clustering model accuracy by **19%**.
- Integrated predictive customer analytics and behavior models into production microservices via **Flask REST APIs**, serving real-time scoring for **200K+ daily user requests** with **sub-150ms latency** during peak-traffic hours.
- Collaborated with database engineering teams to design high-performance data queries using **SQL**, minimizing extraction delays and reducing model data-retrieval bottlenecks by **35%** during batch prediction runs.
- Built automated statistical validation scripts to evaluate predictive accuracy across 3 production marketing models prior to deployment, ensuring stable performance and preventing prediction errors.
- Optimized memory utilization for heavy linear algebra computations within the customer segmentation pipeline, slashing server computational overhead by **25%** and reducing data processing costs.

EDUCATION

Masters of Science in Data Analytics | Indiana Wesleyan University

June 2026 -Current

Master of Science in Computer and Information Sciences | University of North Texas, Texas

January 2022 - May 2023